SARRTHI: Software Requirements Specification

Draft v0.1 (Sections 1 to 5) | Week 5 Deliverable | 7 August 2026

Project: SARRTHI, a Proactive Compliance and Payment-Monitoring Platform for MSME Suppliers to Government Transport Bodies

Backend technology mandate: Java (Spring Boot)

Academic Project — Not an Official Government Portal.

Document Control

Item	Detail
Document	Software Requirements Specification (SRS)
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Prepared by	Project team: Member 1 (Team Lead), Member 2, Member 3, Member 4
Based on	Project Plan and Market Research Notes (Week 2); Problem Statement and Feasibility Report (Week 3); Requirement Analysis and Use Cases (Week 4)

Section	Drafted by
Outline, Sections 1 and 2, Module 1 (4.1)	Member 1
Modules 2 and 3 (4.2, 4.3)	Member 2
Modules 4, 5 and 6 (4.4, 4.5, 4.6)	Member 3
Section 3 (architecture), Modules 7 to 9 (4.7 to 4.9), Section 5	Member 4

Version	Date	Change
0.1	7 August 2026	First draft of Sections 1 to 5

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional requirements and the architecture of SARRTHI, a proactive compliance and payment-monitoring platform for Micro, Small and Medium Enterprises (MSMEs) that supply Indian Railways and comparable government transport bodies. It is written for the developers and testers who will build and verify the system, for the Team Lead, and for the project supervisor who will evaluate it. It also fixes the backend technology as Java.

1.2 Scope

SARRTHI will allow an MSME vendor to:

- track every invoice against the statutory 45-day payment window and see the time remaining or overdue;
- estimate the interest it may be owed on delayed payments;
- monitor buyer behaviour and procurement-quota compliance, and see a risk score before bidding;
- keep orders, invoices, delivery proofs and other evidence in one place;
- prepare for escalation toward the government's MSME ODR Portal; and
- see its position in a dashboard and in exportable reports.

SARRTHI is a complement to the MSME ODR Portal, not a replacement for it. It performs no arbitration, conciliation, mediation or legal advice. Interest figures are estimates and not guaranteed recoveries.

1.3 Definitions, Acronyms and Abbreviations (Common Glossary)

The team agreed on this single glossary in Week 5 after terminology differed between sections.

Term	Definition
MSME	Micro, Small and Medium Enterprise
MSMED Act	Micro, Small and Medium Enterprises Development Act, 2006
ODR	Online Dispute Resolution; the Ministry of MSME's ODR Portal
MSEFC	Micro and Small Enterprise Facilitation Council
Udyam	Government registration for MSMEs
GSTIN / PAN	GST identification number / Permanent Account Number
PO	Purchase Order
EMD	Earnest Money Deposit
IREPS / GeM	Indian Railways e-procurement system / Government e-Marketplace
TReDS	Trade Receivables Discounting System
Acceptance date	Date on which the buyer accepts the goods or services; start of the payment clock
Payment clock	The count of days from the start date to the due date, and beyond it
Due date	Last day for payment under the agreed term, never more than 45 days from acceptance
Overdue	Unpaid after the due date

Term	Definition
Monitoring window	The configurable number of days used by the payment clock; 45 by default
Evidence checklist	List of documents a vendor should hold before any filing
Buyer risk score	Rule-based indicator of a buyer's payment reliability
OCR	Optical Character Recognition
RBAC	Role-Based Access Control
JWT	JSON Web Token, used for stateless authentication
REST / API	Representational State Transfer / Application Programming Interface
ORM / JPA	Object-Relational Mapping / Java Persistence API
DTO	Data Transfer Object
DPDP Act	Digital Personal Data Protection Act, 2023

1.4 References

1. MSMED Act, 2006, Sections 15, 16, 18, 20 and 21.
2. Ministry of MSME, Notification S.O. 1364(E), 21 March 2025 (MSME classification).
3. Public Procurement Policy for Micro and Small Enterprises (MSEs) Order, 2012, as amended.
4. MSME ODR Portal (odr.msme.gov.in), launched 27 June 2025.
5. SARRTHI Project Plan and Market Research Notes (Week 2).
6. SARRTHI Problem Statement and Feasibility Report (Week 3).
7. SARRTHI Requirement Analysis and Use Cases (Week 4).

1.5 Conventions

- **shall** marks a mandatory requirement; **should** marks a desirable one; **may** marks an optional one.

- Requirements are numbered **FR-m.n**, where m is the module number.
- **Origin** is Original (part of the initial idea), Added-Necessary (needed for a complete, deployable product) or Advanced (an upgrade of an original feature).
- **Priority** is Must, Should or Could.

2. Overall Description

2.1 Product Perspective

SARRTHI is a new, standalone web application with a mobile-responsive interface. Native mobile apps are a later phase. It runs independently of government systems.

External entity	Relationship	Status
MSME ODR Portal	SARRTHI prepares information the vendor may use when filing; it does not file on the vendor's behalf	Information only
IREPS and GeM	No public developer API identified; not a dependency	Future option
Tally and GST e-invoice	Possible sources of invoice data	Later phase
WhatsApp Business and email	Alert delivery channels	WhatsApp later; email as fallback

2.2 Product Functions

Module	Summary
1. User, Identity and Access	Registration, login, business profile, roles, consent
2. Invoice and Payment Monitoring	Orders, invoices, payments, payment clock, interest estimates, working-capital gap
3. Compliance Intelligence	Quota view, checklists, buyer risk, pre-bid score, tender tracking
4. Communication and Escalation	Alerts and the escalation ladder toward the ODR Portal

Module	Summary
5. Document and Data Capture	Document vault, evidence checklists, OCR and integrations later
6. Financial Support	Exportable reports and a TReDS referral
7. Collective Intelligence	Aggregated, anonymised vendor data; later phase
8. Reporting, Admin and Accessibility	Dashboard, audit trail, security, accessible interface
9. Monetisation	Freemium tiers and a lender-facing API; later phase

2.3 User Classes and Characteristics

User class	Description	Access
MSME Owner	Primary account holder	Full access to own data, billing and consent settings
MSME Staff	Accountant or office staff	Data entry for orders, invoices and documents; no billing or consent settings
Platform Administrator	Internal team	Verification, moderation and support
Trade Body (future)	Industry association	Read-only access to aggregated, anonymised data
Lending Partner (future)	Bank or NBFC	Consent-gated, API-based access

2.4 Operating Environment

- **Backend:** Java 17 LTS or later with the Spring Boot framework.
- **Database:** PostgreSQL in production; an in-memory database may be used for local development.
- **Frontend:** React web application, responsive on desktop and mobile browsers.
- **Hosting:** Containerised with Docker, deployable on any major cloud provider.

2.5 Design and Implementation Constraints

- The backend shall be implemented entirely in Java, so that a small team can maintain a single technology stack.
- The first version shall work on vendor-supplied data because no public IREPS API exists.
- The system shall follow the principles of the DPDP Act, 2023 when handling vendor financial and business data.
- The system shall not perform, or claim to perform, any legal, arbitration or dispute-resolution function.
- The system shall always display the disclaimer 'Academic Project — Not an Official Government Portal.' and shall not use government logos.

2.6 Assumptions and Dependencies

- Vendors will enter or upload their own order, invoice and payment data.
- The acceptance date of goods is known and can be recorded.
- A critical mass of vendors is needed before Module 7 produces meaningful output, so that module is phased later.
- WhatsApp Business API access can be obtained if the notification module is built.

3. System Architecture (Java Backend)

3.1 Architecture Overview

SARRTHI follows a layered, modular-monolith architecture: one Spring Boot application, internally divided into modules that match Section 4. This is faster for a small team to build, test and deploy, and it allows any module to be extracted into its own service later without a redesign.

Layer	Responsibility	Technology
Presentation	Web interface, forms, dashboards, alert display	React, TypeScript, Tailwind CSS
API	REST endpoints used by the frontend and future mobile apps	Spring Web controllers

Layer	Responsibility	Technology
Business logic	Payment clock, interest calculation, risk scoring, alert triggers	Java service classes
Security	Authentication, access control, ownership checks	Spring Security, JWT
Scheduling	Day-30 and day-40 alert jobs and nightly aggregation (later phase)	Spring Scheduler
Persistence	Data access and ORM mapping	Spring Data JPA, Hibernate
Database	Users, orders, invoices, payments, documents, compliance data	PostgreSQL
Integration	External connectors (later phase)	Java REST clients

3.2 Backend Technology Stack

- **Language and runtime:** Java 17 LTS or later.
- **Framework:** Spring Boot with Spring Web, Spring Data JPA, Spring Security and Bean Validation.
- **Build tool:** Maven.
- **Database migration:** Flyway versioned SQL scripts.
- **Authentication:** stateless JWT with role-based access.
- **API style:** RESTful JSON, documented with OpenAPI and Swagger UI.
- **Testing:** JUnit and Mockito for unit and integration tests.
- **OCR (later):** Tesseract through a Java wrapper, or a cloud OCR service.

3.3 Frontend and Supporting Stack

- **Frontend:** React with TypeScript, Vite and Tailwind CSS; tests with Vitest.
- **Data source layer:** a switch between demo data held in the browser and the REST API.
- **Storage:** relational database for records; file storage for uploaded documents.
- **Notifications:** in-app first; email and WhatsApp later.
- **Deployment:** Docker containers with an automated build pipeline.

3.4 Backend Package Structure

Package	Responsibility
controller	REST endpoints; receive requests and return responses
service	Business rules and transactions, for example the payment clock and interest calculation
repository	Spring Data JPA interfaces for database access
entity	JPA classes mapped to database tables
dto	Request and response objects, so entities are never exposed directly
security	JWT creation and validation, authentication filter, security configuration
config	CORS and OpenAPI configuration
exception	Custom exceptions and a global exception handler returning consistent errors
util	Pure helper classes, such as the MSMED date and interest calculator

3.5 Request Flow

1. The browser sends a request with a JWT in the Authorization header.
2. The authentication filter validates the token and identifies the user.
3. The controller validates the input and calls the service.
4. The service applies the business rules and always works on the signed-in user's own records.
5. The repository reads or writes the database.
6. The controller returns a DTO as JSON. Errors pass through the global exception handler.

3.6 Key Design Decisions

- Money is held as exact decimals (BigDecimal in Java, NUMERIC in the database) and never as floating-point numbers.
- Every business record belongs to one user, and every query is filtered by that user.
- Calculation rules are kept in plain utility classes so they can be unit tested without a database.
- Estimates always carry their assumptions and a non-legal-advice label.

4. Functional Requirements

The tables below list every requirement by module. Detailed rules, use cases and worked examples are in the Requirement Analysis document.

4.1 Module 1: User, Identity and Access

ID	Requirement	Origin	Priority
FR-1.1	The system shall let a vendor register with email, password, name and company.	Added-Necessary	Must
FR-1.2	The system shall authenticate users with stateless JWT sessions and support login, logout and session lookup.	Added-Necessary	Must
FR-1.3	The system shall store a business profile with Udyam number, GSTIN and PAN.	Added-Necessary	Must
FR-1.4	The system should verify the Udyam registration before granting access to compliance features.	Added-Necessary	Should
FR-1.5	The system should provide role-based access for Owner and Staff.	Added-Necessary	Should
FR-1.6	The system shall ensure that a user can read and change only their own records.	Added-Necessary	Must
FR-1.7	The system shall let a user change their password.	Added-Necessary	Should
FR-1.8	The system shall record explicit consent and privacy preferences, including consent for any data sharing.	Added-Necessary	Must

4.2 Module 2: Invoice and Payment Monitoring (Core)

ID	Requirement	Origin	Priority
FR-2.1	The system shall let a user create, view, edit and delete invoices with number, buyer, amount, dates, terms and an optional linked order.	Added-Necessary	Must
FR-2.2	The system shall show a payment clock for each invoice: due date, days remaining or overdue, and status.	Original	Must
FR-2.3	The system shall let a user record part and full payments and shall recalculate the balance and status.	Added-Necessary	Must

ID	Requirement	Origin	Priority
FR-2.4	The system shall make the monitoring window configurable and shall display the assumptions behind each result.	Original	Must
FR-2.5	The system shall estimate interest on overdue invoices, both as statutory compound interest and as a simple estimate at a user-set rate.	Original	Must
FR-2.6	The system should provide an interest calculator for what-if amounts and rates.	Added-Necessary	Should
FR-2.7	The system should let a user search, filter and export the invoice list.	Added-Necessary	Should
FR-2.8	The system shall let a user manage purchase orders, delivery and acceptance dates.	Added-Necessary	Must
FR-2.9	The system should check that an invoice or order carries the fields a future ODR filing would need.	Original	Should
FR-2.10	The system should estimate the working-capital gap from pending invoices.	Original	Should
FR-2.11	The system may scan a PO before acceptance and flag a missing written payment term.	Original	Could

4.3 Module 3: Compliance Intelligence

ID	Requirement	Origin	Priority
FR-3.1	The system should estimate whether a buying zone meets the 25% MSME procurement target (configurable).	Original	Should
FR-3.2	The system should provide a compliance checklist that a user can tick off.	Added-Necessary	Should
FR-3.3	The system should show each buyer's overdue invoices and average delay.	Original	Should
FR-3.4	The system should show a rule-based pre-bid risk score at the point of bidding.	Original	Should

ID	Requirement	Origin	Priority
FR-3.5	The system should let a user track tenders, their MSME eligibility and saved tenders.	Added-Necessary	Should
FR-3.6	The system may flag Railway Board circulars relevant to a vendor's product categories.	Original	Could
FR-3.7	The system may let a vendor rate a buyer after contract closure.	Added-Necessary	Could

The quota dashboard (FR-3.1) is the single-vendor version of the crowd-sourced quota tracker (FR-7.1). One underlying service shall be built, and crowd-sourced data will later improve its accuracy.

4.4 Module 4: Communication and Escalation

ID	Requirement	Origin	Priority
FR-4.1	The system shall alert the user to upcoming and overdue deadlines in the application.	Original	Must
FR-4.2	The system should send day-30 and day-40 reminders by WhatsApp or email.	Original	Should
FR-4.3	The system should provide an escalation ladder: reminder, formal notice citing Section 15, then an ODR-ready package.	Original	Should
FR-4.4	The system may pre-fill the data required for an ODR filing from tracked records.	Original	Could
FR-4.5	The system may provide a rights chatbot and a full multi-language interface.	Original	Could

4.5 Module 5: Document and Data Capture

ID	Requirement	Origin	Priority
FR-5.1	The system shall provide a document vault with upload, list, download and delete, organised by category and linked to orders and invoices.	Original	Must

ID	Requirement	Origin	Priority
FR-5.2	The system should provide an evidence checklist for each order.	Original	Should
FR-5.3	The system may capture invoice data from a phone camera using OCR.	Added-Necessary	Could
FR-5.4	The system may import invoice data from Tally and from GST e-invoice records.	Added-Necessary	Could

4.6 Module 6: Financial Support Layer

ID	Requirement	Origin	Priority
FR-6.1	The system should export invoices, orders and buyer delays to CSV.	Added-Necessary	Should
FR-6.2	The system may provide print-friendly reports and PDF or Excel export.	Added-Necessary	Could
FR-6.3	The system may refer a vendor to TReDS financiers once an invoice is overdue past a threshold.	Original	Could

4.7 Module 7: Collective Intelligence (Network Layer)

ID	Requirement	Origin	Priority
FR-7.1	The system may aggregate anonymised vendor data to estimate a zone's real MSME-spend share.	Advanced	Could
FR-7.2	The system may warn all vendors in a product category when one reports a specification-change rejection.	Advanced	Could
FR-7.3	The system may bundle matching complaints from 10 or more vendors into one representation.	Original	Could
FR-7.4	The system may forward aggregated, anonymised non-compliance data to trade bodies.	Original	Could

4.8 Module 8: Reporting, Admin and Accessibility

ID	Requirement	Origin	Priority
FR-8.1	The system shall provide a dashboard with KPIs, monthly financials, invoice status, alerts, buyer risk and an activity feed.	Added-Necessary	Must
FR-8.2	The system shall provide a responsive, accessible interface.	Added-Necessary	Must
FR-8.3	The system shall store passwords as hashes and protect traffic with TLS in deployment.	Added-Necessary	Must
FR-8.4	The system should keep an audit trail of significant actions and allow a clean compliance history to be exported.	Added-Necessary	Should
FR-8.5	The system should provide a demo mode with realistic sample data and a reset function.	Added-Necessary	Should
FR-8.6	The system may provide an admin dashboard and an offline-first mode for poor connectivity.	Added-Necessary	Could

4.9 Module 9: Monetisation and External Revenue

ID	Requirement	Origin	Priority
FR-9.1	The system may manage freemium and paid tiers.	Added-Necessary	Could
FR-9.2	The system may expose a consent-gated API of a vendor's compliance history to lenders.	Advanced	Could

5. Advanced and Future Features

These are not required for the first working version. They are high-value upgrades once the core platform is proven.

Feature	Description	Prerequisite
AI payment-default prediction	A model trained on historical buyer payment patterns to predict default risk	Enough historical data
Predictive cash-flow analytics	Forecast of a vendor's cash position weeks ahead	Invoice, payment and seasonal history
Voice-based vernacular assistant	A voice version of the rights chatbot for lower-literacy users	Chatbot (FR-4.5)
Blockchain document integrity proof	Tamper-evident record of when a document was received	Document vault (FR-5.1)
Direct government data integration	Verified official figures replace crowd-sourced estimates if IREPS or GeM offers a public API	Public API
Native mobile applications	Android and iOS apps with better offline support and push notifications	Validated web platform

Open Issues for Week 6

ID	Issue	Status
OI-1	Terminology differed between sections	Resolved: common glossary in Section 1.3
OI-2	Overlap between the quota dashboard and the crowd-sourced quota tracker	Resolved: one service, with crowd-sourced data as a later enhancement (note under 4.3)
OI-3	Procurement quota figure: early notes said 20%, the current policy says 25%	Resolved: 25% by default, configurable
OI-4	Statutory compound interest versus a simple estimate	Both offered, with assumptions shown (FR-2.5)

ID	Issue	Status
OI-5	Section 6 (non-functional requirements), Section 7 (phasing matrix) and Section 8 (technology summary)	Pending: to be written in Week 6
OI-6	Minimum Java version	Java 17 LTS or later; the team confirms the exact version at project setup